

SETUHAUL LOGISTICS

Driver Exception and Dock Slot Coordination

An open-ended FDE classroom problem

FDE classroom case | August 2026

The central challenge

A driver who may miss a warehouse appointment needs help through chat. The difficult part is not answering one driver. The difficult part is handling many drivers at the same time when several of them need the same limited set of feasible receiving slots.

What this brief contains

- A plain-English explanation of freight, warehouses, docks and appointments
- A system and data primer before the technical problem begins
- The reason an AI agent is useful — and the decisions it should not make
- Driver-facing conversation and simultaneous-capacity challenges
- A realistic data package, open design questions and an optional facility-level scheduling extension

Important: This document describes the business problem and the available business data. It does not prescribe an agent framework, tool set, storage design, concurrency mechanism, allocation algorithm or deployment pattern.

1. • Company and Business Context

1.1 The company

SetuHaul Logistics Pvt. Ltd. is a mid-sized third-party logistics company operating full-truckload road freight across North and West India. It coordinates company-owned and contracted trucks carrying domestic shipments to factories, warehouses and distribution centres.

The operating figures below define SetuHaul's current scale for this problem brief.

CLASSROOM OPERATING PROFILE	OPERATING VALUE
Loads coordinated each day	280-360
Destination facilities	6

CLASSROOM OPERATING PROFILE	OPERATING VALUE
Dock doors across all facilities	32
Scheduled receiving appointments per day	190-240
Driver exception messages on a normal day	15-25
Messages during a disruption spike	20-35 within 30 minutes
Operations coordinators per shift	5

1.2 Why delays matter

A freight plan is a chain of commitments: the right truck, load, route, arrival time, warehouse capacity, unloading labour and stakeholder communication. A late arrival can break several commitments together.

External scale reference. ATRI reported that drivers experienced detention at 39.3% of stops in 2023. It estimated more than 135 million annual detention hours, 72 million gallons of diesel wasted while idling and \$11.5 billion in productivity losses. These are U.S. industry figures used only to show the scale of waiting and coordination problems.

One delay is manageable. A wave of delays is not. Rain, congestion, road closures or loading backlogs can create many incomplete driver messages while the number of usable replacement slots remains limited.

2. Freight Operations in Plain English

2.1 Follow one truck from origin to delivery

STAGE	WHAT HAPPENS	BUSINESS DATA CREATED OR UPDATED
1. Shipment planned	A load is created with origin, destination, goods and required delivery time.	shipment_id, origin, destination, product class, priority
2. Driver and vehicle assigned	A carrier assigns a driver and truck to move the load.	driver_id, vehicle_id, carrier_id
3. Warehouse appointment booked	The destination reserves a receiving window for unloading.	facility_id, dock_id, slot_id, appointment_id
4. Truck travels	The driver moves toward the destination. The original ETA remains in the plan, and the	planned_eta, latest_declared_eta, eta_updated_at, shipment status

STAGE	WHAT HAPPENS	BUSINESS DATA CREATED OR UPDATED
	driver may declare a revised ETA when conditions change.	
5. Gate and yard arrival	The truck checks in and may wait inside the facility yard.	actual_gate_in_at, arrival_status, queue_status
6. Dock and unload	The truck backs into a loading bay, goods are unloaded, and the visit is completed.	dock-in, unload start/end, proof of delivery

Simple movement: Origin warehouse → truck in transit → destination gate → yard/waiting area → dock door/loading bay → unloading → exit

2.2 Important terms

TERM	PLAIN-ENGLISH MEANING
Facility / warehouse	The destination site that receives goods. It may be a factory, warehouse or distribution centre.
Gate	The controlled entrance where the driver and vehicle are checked in.
Yard	The open area inside or near the facility where trucks wait before being called to a dock.
Dock door / loading bay	The physical position where a truck reverses against the warehouse building so goods can be loaded or unloaded. “Dock station” is not the preferred term in this brief.
Dock appointment	A booking that says a particular shipment is expected during a defined receiving window.
Appointment slot	The start and end time available for a dock or a unit of receiving capacity, for example 7:30-8:00 PM.
Compatible dock	A dock that can physically and operationally handle the truck and goods — for example vehicle length, refrigeration, product type or required equipment.
ETA	Estimated time of arrival. The first ETA comes from the shipment plan. If conditions change, the driver may declare a revised ETA. This exercise does not require live GPS tracking.

TERM	PLAIN-ENGLISH MEANING
Exception	An event such as a breakdown, traffic delay or late departure that makes the original plan uncertain.
Detention	Time a driver spends waiting at a facility beyond the expected loading or unloading period.

3. · Where the Data Comes From

A driver message does not contain the complete answer. Different systems hold different parts of the business truth. Students should understand these systems before designing any agent or workflow.

BUSINESS SYSTEM	WHAT IT KNOWS	CLASSROOM DATA
TMS — Transportation Management System	Shipment, driver, vehicle, origin, destination, original plan, planned ETA and shipment status.	drivers, vehicles, shipments
Dock scheduler / WMS	Facilities, dock doors, receiving windows, existing bookings, operating rules and capacity.	facilities, docks, appointment_slots, appointments, facility_rules
Gate / yard check-in	Whether a truck has actually arrived, when it entered, and whether it is waiting, docked or completed.	facility_checkins
Driver chat and ETA updates	What happened, the latest ETA declared by the driver, questions, preferences and constraints.	driver_exceptions, eta_updates, chat_messages
Contact / messaging system	Warehouse, customer and carrier contacts plus confirmations or replies.	contacts, operational_messages

Why integration matters

The shipment plan provides the original ETA. The driver can declare a revised ETA. Gate check-in establishes actual arrival. The warehouse system knows dock capacity, while messaging carries confirmations. No single system contains the full operational answer.

3.1 The basic data relationships

RELATIONSHIP	MEANING
driver → shipment	Which active load the person in chat is currently moving.
shipment → vehicle	Which truck must be checked for dock compatibility.
shipment → destination facility	Which warehouse schedule and local rules apply.
shipment → eta_updates	The original planned ETA and the latest ETA explicitly declared by the driver.
shipment → facility_checkins	Whether the truck has arrived and whether it is waiting at the gate, yard or dock.
facility → dock → slot	Which physical receiving capacity is available at a specific time.
shipment → appointment → slot	Which receiving window is currently planned for the load.
exception → thread → messages	Which conversation belongs to which reported operational problem.

4. • One Example Across the Data

Use one connected example when introducing the tables. Students should be able to trace a single driver message into the relevant rows before they think about tools or agents.

Driver message

“Tyre damaged near Neemrana. Repair may take 45 minutes. Can I get a slot after 7 PM? I must leave before 9 PM for another pickup.”

TABLE	EXAMPLE ROW	WHAT THE ROW TELLS US
drivers	DRV-027 Ravi Kumar carrier CAR-08	Who is speaking and which carrier is responsible.
vehicles	VEH-031 32-foot dry van active	The truck type that a dock must support.
shipments	SHP-1042 DRV-027 VEH-031 destination FAC-JPR-01 planned ETA 17:20	The active movement, destination and original expected arrival.
eta_updates	ETA-880 SHP-1042 driver revised ETA 19:10 declared 16:08	The latest arrival time explicitly declared by the driver.

TABLE	EXAMPLE ROW	WHAT THE ROW TELLS US
facilities	FAC-JPR-01 Jaipur DC open 06:00-23:00	The warehouse, local operating hours and location.
docks	DOCK-04 FAC-JPR-01 dry freight max 32-foot	A physical loading bay that may be compatible.
appointment_slots	SLOT-1938 DOCK-04 17:30-18:00 open	A unit of receiving time and capacity.
appointments	APT-552 SHP-1042 SLOT-1938 confirmed	The driver's original booked warehouse window.
facility_checkins	No row yet for SHP-1042	The truck has not reached the destination gate. Once it arrives, actual gate-in and queue status become available.
driver_exceptions	EXC-778 SHP-1042 tyre issue reported 16:05	The operational problem being handled.
chat_messages	MSG-9001 EXC-778 driver free-text message	The exact conversational input and its thread.
facility_rules	FAC-JPR-01 dry vans up to 32 ft effective all day	A local eligibility rule that affects feasible choices.

4.1 Questions the data must answer

- Which active shipment does DRV-027 mean?
- What is the current appointment and when does it become infeasible?
- What is the revised ETA, and how uncertain is it?
- Has the truck already arrived, and is it waiting at the gate, yard or dock?
- Which docks can accept VEH-031 and this product class?
- Which later slots are still usable after considering other appointments and requests?
- Has a slot merely been shown, temporarily held, requested or confirmed?

The example values are illustrative. The final student dataset should contain many linked records, imperfect data and simultaneous requests.

5. Current Operating Reality

5.1 Parties Involved

PARTY	WHAT IT KNOWS	WHAT IT NEEDS DURING AN EXCEPTION
Driver	Physical location, breakdown, traffic and likely delay.	Clear choices, instructions and confirmation.
Dispatcher / carrier	Vehicle assignment and driver availability.	A revised plan and escalation status.
SetuHaul operations	Shipment plan, customer commitment and open exceptions.	A consistent view across many simultaneous requests.
Warehouse planner	Dock capacity, labour, equipment and local rules.	Feasible arrivals without overbooking.
Consignee / customer	Inventory or production dependency.	Reliable revised arrival information.

5.2 What happens today

1. A driver sends a short message such as: “Tyre damaged near Neemrana. Around 90 minutes late.”
2. An operations coordinator identifies the driver, active shipment, destination and existing appointment.
3. The coordinator checks whether the current appointment is still possible.
4. The coordinator checks the facility schedule, local rules and other pending requests.
5. The coordinator discusses or books a revised plan and informs the required parties.
6. Replies, cancellations and changing ETAs can force the plan to be reconsidered.

Why a small message creates a large workflow

The message is unstructured, but the decision depends on shipment context, the latest declared ETA, actual arrival status, dock capacity, facility rules, competing requests and confirmation status. No single message or system contains the complete answer.

5.3 What cannot be guessed

QUESTION	WHY IT CANNOT BE GUESSED
Which shipment is affected?	A driver can have different assignments on different days.
Is the stated delay the full impact?	Repair time and revised arrival time are not always the same.

QUESTION	WHY IT CANNOT BE GUESSED
What is the latest ETA?	The original plan may be stale, and the driver may correct the ETA over several messages.
Has the truck already arrived?	Actual gate-in and queue status come from facility operations, not from the original appointment.
Which slots are feasible?	Availability and eligibility differ by load, vehicle and time.
Is a slot still available?	Another conversation may be trying to claim it at the same time.
Has the warehouse confirmed?	A proposed change and a committed appointment are not the same.

6. · Why Use an AI Agent Here?

SetuHaul does not need an LLM to calculate warehouse capacity or safely allocate scarce slots. Those are structured operational decisions. The AI is useful because the work begins and ends through messy, multi-turn human conversation.

6.1 Why a fixed form is not enough

A form works well when every driver knows the shipment number, delay, revised ETA and desired action. Real messages are often incomplete, informal and corrected over several turns.

DRIVER SAYS	WHAT MUST BE UNDERSTOOD OR CLARIFIED
"I will be around two hours late."	Which shipment, why, and whether two hours is the total ETA impact.
"Anything after 7?"	Which facility, compatible options, and whether the driver is asking to view or book.
"The second one works, but I need to leave by 9."	Which previously shown option and whether unloading can finish before the next commitment.
"Has it been confirmed?"	The current status of the same exception thread and latest warehouse response.

6.2 What the AI layer contributes

- Understands free-text driver messages and informal wording.
- Asks only for information that is missing or ambiguous.
- Maintains context across follow-up questions and corrections.
- Connects conversational intent to structured business data and controlled actions.
- Explains available choices, constraints and status in simple language.

- Continues the same thread when the driver returns later for an update.

6.3 What the AI should not decide by itself

- Whether two drivers can be promised the same capacity.
- Whether a vehicle is physically compatible with a dock.
- How scarce capacity is prioritised when business rules conflict.
- Whether a booking is committed successfully in the system of record.
- Driver-safety, legal, penalty or exceptional commercial decisions.

The FDE lesson

The AI agent is the conversational coordination layer. Students must still design a trustworthy operational layer for feasibility, freshness, capacity, concurrent requests and confirmation. The brief intentionally leaves that design open.

7. • The Problem Students Must Solve

Problem statement

How might SetuHaul provide a conversational way for drivers to report delays, ask questions and consider revised appointments, while ensuring that many simultaneous requests are handled against limited warehouse capacity without infeasible or conflicting promises?

7.1 First-level challenge: one driver

- Understand what happened and what information is missing.
- Connect the conversation to the correct active shipment and original appointment.
- Help the driver understand whether the original plan still works.
- Show suitable possibilities and answer follow-up questions.
- Keep the driver informed about whether a change is proposed, pending or confirmed.

7.2 The real challenge: many drivers

The system may receive many requests at once. Five drivers may ask for a 6:00 PM window when only one compatible dock is free. Two conversations may read the same availability before either completes. A cancellation may free a slot while another driver is deciding. A high-priority load may appear after lower-priority choices have already been discussed.

CONCURRENT EVENT	BUSINESS RISK
Two drivers choose the same slot within seconds.	Double booking or a broken promise.
A shown slot disappears before confirmation.	The driver receives stale guidance.
A high-priority load arrives after earlier requests.	Unclear fairness or service-level breach.
A driver changes ETA during the conversation.	Previously discussed options may become infeasible.
A warehouse blocks one dock unexpectedly.	Several appointments may require reconsideration.
A driver retries or sends duplicate messages.	Duplicate requests or repeated actions.

The allocation question

The challenge is not simply “find an empty row.” Students must decide what availability means while conversations are active, how requests compete, when a promise becomes binding, and what happens when demand is greater than capacity.

7.3 Optional extension: a scheduling engine as a tool

The conversational agent handles one driver’s message and questions. Students may optionally design a separate scheduling engine that considers all relevant trucks for one facility together. The agent can call this engine as a controlled tool. The engine receives structured operational data and returns a proposed schedule or ranked feasible options; it does not interpret free text.

Data boundary

This exercise does not require live GPS tracking. The scheduling engine uses the original planned ETA, the latest ETA explicitly declared by the driver, and actual gate-in time once a truck reaches the facility.

Example: several trucks compete for two dock doors

At 5:25 PM, the Jaipur facility has two dock doors. The latest operational snapshot looks like this:

TRUCK OR FACILITY STATE	LATEST KNOWN OPERATING DATA
SHP-201 — arrived early	Booked for 5:30 PM; gate-in at 5:05 PM; waiting in the yard; 40-minute unload; compatible with D1 or D2.
SHP-202 — arrived late	Booked for 5:00 PM; gate-in at 5:25 PM; waiting at the gate; 30-minute unload; compatible only with D2.
SHP-203 — expected late	Booked for 5:45 PM; not yet arrived; driver updated ETA to 6:35 PM; 45-minute unload; compatible with D1 or D2.
SHP-204 — currently unloading	Already on D1; expected to finish at 5:40 PM. This work cannot simply be moved.
Facility capacity	D1 is occupied until about 5:40 PM. D2 is free now. Operating hours continue until 11:00 PM.

The question is no longer only “find a later slot for SHP-203.” The facility must decide what to do with the early truck, the late truck already waiting, the truck expected much later, and the work already in progress. A scheduling tool may propose a facility-wide sequence while respecting commitments and capacity.

How the business data maps to a scheduling model

BUSINESS CONCEPT	SCHEDULING-MODEL INTERPRETATION
Truck / shipment	A job that requires service.
Dock door	A machine or limited resource.
Original or latest driver-declared ETA	A release time: the truck should not be scheduled before it can arrive.
Actual gate-in	Evidence that the truck is available now and may already be waiting.
Expected unloading duration	Processing time required on the dock.
Booked appointment	A promised time window, due time or service commitment.
Vehicle and product compatibility	The set of dock doors that are allowed for that truck.
Shipment priority	A weight or penalty when waiting or lateness is calculated.
Truck already unloading or a protected commitment	A fixed task or constraint that the new schedule should not move.

What students may explore

- A simple dispatching baseline such as appointment-first, first-arrived-first-served, earliest due time or shortest unloading time.
- A transparent score-based heuristic combining waiting time, lateness, priority and dock compatibility.
- Constraint programming or mixed-integer optimisation for assigning trucks to dock-time intervals.
- Rolling-horizon scheduling that recalculates when a driver updates ETA, a truck checks in, unloading completes, a slot is cancelled or a dock becomes unavailable.
- The objective itself: minimise total waiting, appointment lateness, overtime, priority violations or unnecessary changes to an already communicated schedule.

This is optional. A student team can keep the scheduling tool rule-based, provided the rules are explicit and the team demonstrates what happens when several trucks compete. The optional engine is facility-level; national route or fleet optimisation remains outside the assignment.

Optional operational-research references

1. Google OR-Tools — Scheduling overview. A practical introduction to resource-constrained scheduling.
2. Google OR-Tools — The Job Shop Problem. Useful for thinking of trucks as jobs and dock doors as machines.
3. Monemia and Gelareh — Dock Assignment and Truck Scheduling Problem. A research example focused directly on dock assignment, truck scheduling and resource constraints.

8. • Make the Experience Genuinely Chat-Based

The driver should be able to report an exception, ask practical questions, compare options, add personal operating constraints, change a choice and return later for status. The conversation is therefore both informational and operational.

CONVERSATION TYPE	EXAMPLE DRIVER MESSAGE
Report	"I am stuck near Neemrana and may be 90 minutes late."
Clarify	"Repair will take 45 minutes, but traffic after that is uncertain."
Ask for options	"What are the next two slots after 6 PM?"
Add a constraint	"I need to leave by 9 PM for my next pickup."
Compare	"Which option has the shortest expected waiting time?"
Facility question	"Does the 7:30 slot accept a 32-foot vehicle?"
Choose	"Take the second option."
Change mind	"Do not book 7:30. Check tomorrow morning."

CONVERSATION TYPE	EXAMPLE DRIVER MESSAGE
Status	"Has the warehouse confirmed my new slot?"
Fallback	"There is no slot today. What should I do next?"

8.1 Conversation identity

CONCEPT	RECOMMENDED CLASSROOM MEANING
User / session identity	The authenticated driver_id.
Conversation thread	One active shipment exception request.
Request identifier	A unique exception_id connected to the thread.
Shipment context	The shipment affected by this exception.
Conversation state	What is known, what remains unclear and the status of any proposed change.

Important interaction rule

Showing an option is not necessarily the same as reserving it, and reserving it is not necessarily the same as confirming it. Students must define these meanings and what the driver sees at every stage.

9. Operating Constraints Students Must Consider

9.1 Slot feasibility

- The driver must be able to reach the facility before the receiving window begins.
- The slot must fall within facility operating hours.
- The dock must support the vehicle type and load requirements.
- The unloading duration must fit the remaining capacity.
- The facility may restrict carrier, product class, temperature control or appointment type.
- Availability may change between viewing, discussing, requesting and confirming a slot.

9.2 Allocation policy

SetuHaul has not defined one universal allocation policy. Students must expose the trade-offs and propose a defensible policy rather than hiding the decision inside a prompt.

POSSIBLE CONSIDERATION	WHY IT MAY MATTER
First confirmed, first served	Simple and predictable, but may ignore service criticality.
Customer or shipment priority	Protects important commitments but may appear unfair.
Perishable or time-sensitive goods	Delay may create greater operational or product risk.
Earliest feasible arrival	May reduce waiting and improve dock utilisation.
Driver duty time and next assignment	The driver may have limited operating time remaining.
Fairness across carriers	Prevents one carrier from repeatedly consuming scarce capacity.
Warehouse utilisation and overtime	Some choices may create labour or equipment cost.

9.3 Human control

- Driver safety decisions remain with the driver, carrier and human operations team.
- Commercial penalties, compensation and customer commitments require authorised approval.
- No-feasible-slot cases must support escalation rather than inventing an answer.
- Contradictory information, regulated loads and emergency situations require manual takeover.

10. • Core Data Package for Students

Provide the business truth required to understand and design the system, but do not pre-create solution-specific structures such as a slot-hold table, allocation-decision table or agent-actions table. Students should decide whether such structures are necessary.

10.1 Identity and movement data

TABLE	BUSINESS MEANING	IMPORTANT FIELDS
drivers	The person using chat and their carrier context.	driver_id, name, phone, carrier_id, status, home_base
vehicles	The truck assigned to a movement and its physical requirements.	vehicle_id, carrier_id, vehicle_type, length_ft, refrigeration_required, status
shipments	The active freight movement connecting driver, vehicle, origin and destination.	shipment_id, driver_id, vehicle_id, origin_id, destination_id, product_class, priority,

TABLE	BUSINESS MEANING	IMPORTANT FIELDS
		planned_eta, expected_unload_minutes, status
eta_updates	A history of revised ETAs explicitly declared by the driver or operations team. No live tracking is required.	eta_update_id, shipment_id, declared_eta, source_type, declared_at, confidence_note
facility_checkins	Observed arrival and waiting state once the truck reaches the destination facility.	checkin_id, shipment_id, facility_id, gate_in_at, arrival_status, queue_status, dock_in_at, completed_at

10.2 Warehouse capacity data

TABLE	BUSINESS MEANING	IMPORTANT FIELDS
facilities	Receiving sites and their operating calendars.	facility_id, name, city, timezone, open_time, close_time, contact_id
docks	Physical loading bays and compatibility constraints.	dock_id, facility_id, dock_name, supported_vehicle_type, supported_product_class, active_flag
appointment_slots	Units of receiving time and capacity.	slot_id, facility_id, dock_id, start_time, end_time, capacity_units, slot_status
appointments	The current or historical booking connecting a shipment to a slot.	appointment_id, shipment_id, slot_id, status, booked_at, confirmed_at, cancelled_at
facility_rules	Local rules that change eligibility or operating behaviour.	rule_id, facility_id, rule_type, rule_value, effective_from, effective_to

10.3 Exception and conversation data

TABLE	BUSINESS MEANING	IMPORTANT FIELDS
driver_exceptions	Each disruption reported for a shipment.	exception_id, driver_id, shipment_id, exception_type, reported_delay_minutes, latest_declared_eta, reported_at, status
chat_messages	The conversational record for the exception thread.	message_id, thread_id, exception_id, sender_type, message_text, created_at

TABLE	BUSINESS MEANING	IMPORTANT FIELDS
contacts	Operational stakeholders who may need updates or approval.	contact_id, party_type, name, email, phone, facility_id, shipment_id

10.4 Optional enrichment data

TABLE	WHY IT HELPS
facility_capacity_changes	Creates events such as a dock closure, equipment failure or reduced labour.
appointment_history	Shows earlier reschedules, cancellations and changes made to the booked plan.
customer_commitments	Adds service-level and priority context without exposing commercial detail.
operational_messages	Represents warehouse or customer replies outside the driver chat.
scheduling_runs	Optionally stores the input snapshot, proposed sequence, objective values and explanation from a scheduling tool for later review.

10.5 What not to give students in advance

- A completed slot-allocation or scheduling algorithm. Students may design one, but it should not be supplied as the answer key.
- A predefined agent tool list or orchestration graph.
- A ready-made slot-hold or locking model.
- A final priority policy.
- A table that records the “correct” solution for each request.

Why this matters

Students need enough data to understand the business, but the data model should not reveal the intended implementation. Their job is to discover what additional state, controls and interfaces are needed.

11. · How to Make the Dataset Realistic

11.1 Recommended classroom size

ENTITY	SUGGESTED VOLUME
Drivers	80-120
Vehicles	90-140
Shipments	600-1,000 across seven days
ETA updates	800-1,500, including unchanged, corrected and uncertain estimates
Facility check-ins	400-700 gate, yard, dock and completion events
Facilities	6
Docks	24-32
Appointment slots	2,000-3,000
Appointments	900-1,500
Exceptions	250-400
Chat messages	1,500-3,000

11.2 Required stress scenarios

- At least 10 drivers request alternatives for the same facility and evening window while only 3-4 compatible slots exist.
- At the same time, one truck is early and waiting, one truck arrived late and is waiting, one truck is currently unloading, and another driver has declared a later ETA but has not yet arrived.
- Two drivers select the same option within a few seconds.
- One facility reduces capacity after options have already been discussed.
- A cancellation creates a new slot during an active conversation.
- A driver sends duplicate messages because of weak connectivity.
- A driver has more than one shipment record, requiring disambiguation.
- A stated 90-minute repair delay does not equal a 90-minute ETA shift.
- One shipment has higher priority but enters the queue later.
- One request has no feasible same-day slot.
- A warehouse reply conflicts with the stored schedule.

11.3 Data imperfections

- Missing delay duration or uncertain repair completion time.

- Free-text location names and inconsistent spelling.
- A stale latest-declared ETA, a missing ETA timestamp or several corrections within the same conversation.
- Cancelled appointments still visible in history.
- Facility rules effective only during part of the day.
- Different descriptions for the same exception reason.

12. • Open-Ended Student Brief

Students are expected to design and demonstrate a credible solution to the business problem. The brief intentionally does not prescribe the architecture.

12.1 Questions students must answer

1. What information must be collected before useful options can be shown?
2. How is a conversation connected to the correct driver, shipment and appointment?
3. How is revised arrival time determined and uncertainty communicated?
4. What makes a slot feasible for a specific shipment?
5. What does “available” mean while another driver is considering the same slot?
6. At what point does an option become a hold, request, reservation or confirmed booking?
7. How are simultaneous requests ordered when capacity is insufficient?
8. Optional scheduling extension: when should a facility-wide schedule be recalculated, which work is fixed, and what objective is being optimised?
9. How are stale options, cancellations, duplicate messages and retries handled?
10. What happens when there is no feasible slot?
11. What is explained when the preferred slot is not granted?
12. Which decisions require human approval or takeover?
13. How will the team prove that the system did not double-book capacity?

12.2 Expected demonstration

- A driver reports a delay and answers one or more clarification questions.
- The driver asks for later possibilities and compares alternatives.
- Several requests are processed against the same facility schedule.
- Optional: a scheduling tool receives the facility snapshot and proposes a revised sequence after an ETA update or gate check-in.
- At least two requests compete for the same capacity.
- The driver is shown what happens when an option changes or disappears.
- At least one case ends in escalation because no safe or feasible automated outcome exists.

12.3 Out of scope

- National transport-network optimisation; the optional extension is limited to dock scheduling within one facility or a small facility set.
- Carrier selection and freight-rate negotiation
- Autonomous driver-safety decisions
- Customs, hazardous-material and legal-compliance workflows
- Commercial penalty approval

13. • Business Success and Reference

13.1 Business success definition

Success is not “the chatbot answered.” Success means that a driver exception becomes a feasible, current and clearly communicated operating plan without creating a conflict for another driver.

MEASURE	WHAT IT REVEALS
Time from first message to usable outcome	Operational speed
Share resolved without manual takeover	Automation coverage
Conflicting or duplicate allocations	Correctness under concurrency
Options later found infeasible	Freshness and decision quality
No-feasible-slot escalations handled correctly	Safety of failure behaviour
Average driver waiting after rescheduling	Operational outcome
Warehouse slot utilisation	Capacity efficiency
Priority-policy violations	Business-policy compliance
Driver clarification turns	Conversation effort and data completeness

13.2 Industry reference

Opendock is a useful industry reference, not the answer key. Its public materials describe carrier self-scheduling, driver-to-facility communication, digital check-in and real-time appointment syncing. This validates the business category while leaving students to design their own classroom solution.

Sources

1. American Transportation Research Institute (ATRI), "New Research Documents Substantial Financial and Safety Impacts from Truck Driver Detention," September 2024.
2. Opendock, Dock Scheduling Software and Product Tours.
3. Google OR-Tools, Scheduling Overview and The Job Shop Problem.
4. R. N. Monemia and S. Gelareh, Dock Assignment and Truck Scheduling Problem; Consideration of Multiple Scenarios with Resource Allocation Constraints, 2023.
5. R. Mahes, M. Mandjes, M. Boon and P. Taylor, Dynamic Appointment Scheduling, 2021.

Final student challenge

Design a conversational freight-exception service that remains correct when several drivers ask for the same limited capacity at the same time. Optionally, add a facility-level scheduling tool that considers early, late, waiting and future-arriving trucks together. Explain not only what the system says, but how the business can trust the allocation and communication around it.

Note

This project is optional for advanced learners. However, we recommend joining the session and participating in the project building.